	LDM-1	*LDM-2*	*LDM-4*	*LDM-8*	*LDM-16*	*LDM-32*
z-shape	$256 \times 256 \times 3$	$128 \times 128 \times 2$	$64 \times 64 \times 3$	$32 \times 32 \times 4$	$16 \times 16 \times 8$	$88 \times 8 \times 32$
$\lvert\mathcal{Z}\rvert$	-	2048	8192	16384	16384	16384
Diffusion steps	1000	1000	1000	1000	1000	1000
Noise Schedule	linear	linear	linear	linear	linear	linear
Model Size	270M	265M	274M	258M	260M	258M
Channels	192	192	224	256	256	256
Depth	2	2	2	2	2	2
Channel Multiplier	1,1,2,2,4,4	1,2,2,4,4	1,2,3,4	1,2,4	1,2,4	1,2,4
Attention resolutions	32, 16, 8	32, 16, 8	32, 16, 8	32, 16, 8	16, 8, 4	8, 4, 2
Head Channels	32	32	32	32	32	32
Batch Size	9	11	48	96	128	128
Iterations*	500k	500k	500k	500k	500k	500k
Learning Rate	9e-5	1.1e-4	9.6e-5	9.6e-5	1.3e-4	1.3e-4

Table 14. Hyperparameters for the unconditional *LDMs* trained on the CelebA dataset for the analysis in Fig. 7. All models trained on a single NVIDIA A100. *: All models are trained for 500k iterations. If converging earlier, we used the best checkpoint for assessing the provided FID scores.

Task	Text-to-Image	Layout-to-Image		Class-Label-to-Image	Super Resolution	Inpainting	Semantic-Map-to-Image
Dataset	LAION	OpenImages	COCO	ImageNet	ImageNet	Places	Landscapes
f	8	4	8	4	4	4	8
z-shape	$32 \times 32 \times 4$	$64 \times 64 \times 3$	$32 \times 32 \times 4$	$64 \times 64 \times 3$	$64 \times 64 \times 3$	$64 \times 64 \times 3$	$32 \times 32 \times 4$
$\lvert\mathcal{Z}\rvert$	-	8192	16384	8192	8192	8192	16384
Diffusion steps	1000	1000	1000	1000	1000	1000	1000
Noise Schedule	linear	linear	linear	linear	linear	linear	linear
Model Size	1.45B	306M	345M	395M	169M	215M	215M
Channels	320	128	192	192	160	128	128
Depth	2	2	2	2	2	2	2
Channel Multiplier	1,2,4,4	1,2,3,4	1,2,4	1,2,3,5	1,2,2,4	1,4,8	1,4,8
Number of Heads	8	1	1	1	1	1	1
Dropout	-	-	0.1	-	-	-	-
Batch Size	680	24	48	1200	64	128	48
Iterations	390K	4.4M	170K	178K	860K	360K	360K
Learning Rate	1.0e-4	4.8e-5	4.8e-5	1.0e-4	6.4e-5	1.0e-6	4.8e-5
Conditioning	CA	CA	CA	CA	concat	concat	concat
(C)A-resolutions	32, 16, 8	32, 16, 8	32, 16, 8	32, 16, 8	-	-	-
Embedding Dimension	1280	512	512	512	-	-	-
Transformer Depth	1	3	2	1	-	-	-

Table 15. Hyperparameters for the conditional *LDMs* from Sec. 4. All models trained on a single NVIDIA A100 except for the inpainting model which was trained on eight V100.

$$\zeta \leftarrow \text{TokEmb}(y) + \text{PosEmb(y)} \tag{18}$$

$$\text{for } i = 1, \ldots, N:$$

$$\zeta_1 \leftarrow \text{LayerNorm}(\zeta) \tag{19}$$

$$\zeta_2 \leftarrow \text{MultiHeadSelfAttention}(\zeta_1) + \zeta \tag{20}$$

$$\zeta_3 \leftarrow \text{LayerNorm}(\zeta_2) \tag{21}$$

$$\zeta \leftarrow \text{MLP}(\zeta_3) + \zeta_2 \tag{22}$$

$$\zeta \leftarrow \text{LayerNorm}(\zeta) \tag{23}$$

$$\tag{24}$$

With ζ available, the conditioning is mapped into the UNet via the cross-attention mechanism as depicted in Fig. 3. We modify the "ablated UNet" [15] architecture and replace the self-attention layer with a shallow (unmasked) transformer consisting of T blocks with alternating layers of (i) self-attention, (ii) a position-wise MLP and (iii) a cross-attention layer;

see Tab. 16. Note that without (ii) and (iii), this architecture is equivalent to the "ablated UNet".

While it would be possible to increase the representational power of τ_θ by additionally conditioning on the time step t, we do not pursue this choice as it reduces the speed of inference. We leave a more detailed analysis of this modification to future work.

For the text-to-image model, we rely on a publicly available[3] tokenizer [99]. The layout-to-image model discretizes the spatial locations of the bounding boxes and encodes each box as a (l, b, c)-tuple, where l denotes the (discrete) top-left and b the bottom-right position. Class information is contained in c.
See Tab. 17 for the hyperparameters of τ_θ and Tab. 13 for those of the UNet for both of the above tasks.

Note that the class-conditional model as described in Sec. 4.1 is also implemented via cross-attention, where τ_θ is a single learnable embedding layer with a dimensionality of 512, mapping classes y to $\zeta \in \mathbb{R}^{1\times 512}$.

input	$\mathbb{R}^{h\times w\times c}$
LayerNorm	$\mathbb{R}^{h\times w\times c}$
Conv1x1	$\mathbb{R}^{h\times w\times d\cdot n_h}$
Reshape	$\mathbb{R}^{h\cdot w\times d\cdot n_h}$
$\times T$ SelfAttention	$\mathbb{R}^{h\cdot w\times d\cdot n_h}$
$\times T$ MLP	$\mathbb{R}^{h\cdot w\times d\cdot n_h}$
$\times T$ CrossAttention	$\mathbb{R}^{h\cdot w\times d\cdot n_h}$
Reshape	$\mathbb{R}^{h\times w\times d\cdot n_h}$
Conv1x1	$\mathbb{R}^{h\times w\times c}$

Table 16. Architecture of a transformer block as described in Sec. E.2.1, replacing the self-attention layer of the standard "ablated UNet" architecture [15]. Here, n_h denotes the number of attention heads and d the dimensionality per head.

	Text-to-Image	**Layout-to-Image**
seq-length	77	92
depth N	32	16
dim	1280	512

Table 17. Hyperparameters for the experiments with transformer encoders in Sec. 4.3.

E.2.2 Inpainting

For our experiments on image-inpainting in Sec. 4.5, we used the code of [88] to generate synthetic masks. We use a fixed set of 2k validation and 30k testing samples from Places [108]. During training, we use random crops of size 256×256 and evaluate on crops of size 512×512. This follows the training and testing protocol in [88] and reproduces their reported metrics (see † in Tab. 7). We include additional qualitative results of *LDM-4, w/ attn* in Fig. 21 and of *LDM-4, w/o attn, big, w/ ft* in Fig. 22.

E.3. Evaluation Details

This section provides additional details on evaluation for the experiments shown in Sec. 4.

E.3.1 Quantitative Results in Unconditional and Class-Conditional Image Synthesis

We follow common practice and estimate the statistics for calculating the FID-, Precision- and Recall-scores [29,50] shown in Tab. 1 and 10 based on 50k samples from our models and the entire training set of each of the shown datasets. For calculating FID scores we use the `torch-fidelity` package [60]. However, since different data processing pipelines might lead to different results [64], we also evaluate our models with the script provided by Dhariwal and Nichol [15]. We find that results

[3] https://huggingface.co/transformers/model_doc/bert.html#berttokenizerfast

mainly coincide, except for the ImageNet and LSUN-Bedrooms datasets, where we notice slightly varying scores of 7.76 (`torch-fidelity`) vs. 7.77 (Nichol and Dhariwal) and 2.95 vs 3.0. For the future we emphasize the importance of a unified procedure for sample quality assessment. Precision and Recall are also computed by using the script provided by Nichol and Dhariwal.

E.3.2 Text-to-Image Synthesis

Following the evaluation protocol of [66] we compute FID and Inception Score for the Text-to-Image models from Tab. 2 by comparing generated samples with 30000 samples from the validation set of the MS-COCO dataset [51]. FID and Inception Scores are computed with `torch-fidelity`.

E.3.3 Layout-to-Image Synthesis

For assessing the sample quality of our Layout-to-Image models from Tab. 9 on the COCO dataset, we follow common practice [37, 87, 89] and compute FID scores the 2048 unaugmented examples of the COCO Segmentation Challenge split. To obtain better comparability, we use the exact same samples as in [37]. For the OpenImages dataset we similarly follow their protocol and use 2048 center-cropped test images from the validation set.

E.3.4 Super Resolution

We evaluate the super-resolution models on ImageNet following the pipeline suggested in [72], *i.e.* images with a shorter size less than 256 px are removed (both for training and evaluation). On ImageNet, the low-resolution images are produced using bicubic interpolation with anti-aliasing. FIDs are evaluated using `torch-fidelity` [60], and we produce samples on the validation split. For FID scores, we additionally compare to reference features computed on the train split, see Tab. 5 and Tab. 11.

E.3.5 Efficiency Analysis

For efficiency reasons we compute the sample quality metrics plotted in Fig. 6, 17 and 7 based on 5k samples. Therefore, the results might vary from those shown in Tab. 1 and 10. All models have a comparable number of parameters as provided in Tab. 13 and 14. We maximize the learning rates of the individual models such that they still train stably. Therefore, the learning rates slightly vary between different runs *cf*. Tab. 13 and 14.

E.3.6 User Study

For the results of the user study presented in Tab. 4 we followed the protocoll of [72] and and use the 2-alternative force-choice paradigm to assess human preference scores for two distinct tasks. In Task-1 subjects were shown a low resolution/masked image between the corresponding ground truth high resolution/unmasked version and a synthesized image, which was generated by using the middle image as conditioning. For SuperResolution subjects were asked: *'Which of the two images is a better high quality version of the low resolution image in the middle?'*. For Inpainting we asked *'Which of the two images contains more realistic inpainted regions of the image in the middle?'*. In Task-2, humans were similarly shown the low-res/masked version and asked for preference between two corresponding images generated by the two competing methods. As in [72] humans viewed the images for 3 seconds before responding.